MILITARY SPECIFICATION

CARTRIDGE, 20MM, HEI, M56A1

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of electric primed cartridge for use in Guns, Automatic, 20MM, M39 and M61.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–G–2550 — General Specification for Ammunition Except Small Arms Ammunition.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–109 — Inspection Terms and Definitions.

MIL–STD–129 — Marking for Shipment and Storage.

DRAWINGS

ORDNANCE CORPS

D7553814 — Cartridge, High Explosive Incendiary, M56A1.

D7553513 — Electric Primer Resistance Tester M52A3 and M52A5 Cabinet Assembly.

C7258801 — Classification of Defects for Case, Cartridge, 20MM, M103 and M103B1.

FD 20421 — Packing and Marking for Cartridges, 20MM, in Links, in Box, Ammunition, Steel, 20MM Ctg, T46.

FD 20354 — Packing and Marking for Cartridge, 20MM, in Links, In Box, Ammunition, Caliber .50 M2A1, In Wirebound Box.

FD 20353 — Packing and Marking for Cartridge, 20MM, in

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Bulk, in Box, Ammunition, Caliber .50 M2A1, in Wirebound Box.

PUBLICATIONS

ORDNANCE CORPS

ORD–M608–14 — Ammunition Ballistic Acceptance Test Methods, Part VIII.

ORD–M608–11 — Procedures and Tables for Continuous Sampling by Attributes.

GL7555938 — Index to Lists of Inspection Gages and Parts Gaged On: Ctg, HEI, 20MM, M56A1.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions shall be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein.

INTERSTATE COMMERCE COMMISSION

49 CFR 71–78 — Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition — Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71–78 (Rev. 1950).")

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.2 Cartridge assembly. The cartridge and its components shall compy with all requirements of Drawing (Dwg.) D7553814 and with all requirements of the specifications shown thereon.

3.3 Electrical resistance, primed cases. The resistance of the M52A3B1 primer shall be not less than 500 nor more than 5,000,000 ohms when tested as specified in 4.4.1.

3.4 Projectile extraction. No projectile shall separate from the cartridge case at a force less than 1100 pounds or shall fail to separate at a force of 2600 pounds when tested as specified in 4.4.2.

3.5 Velocity. The velocity of the projectile shall be 3380 feet per second (f.p.s.) at 78 feet from the gun muzzle, when tested as specified in 4.4.3.

3.6 Waterproofness. The average velocity of cartridges after submergence in water, as specified in 4.4.3, shall not vary from the average velocity (dry) by more than 100 f.p.s.

3.7 Pressure. The cartridge shall exhibit a pressure of not more than 60,500 lbs. per square inch (p.s.i.) when tested as specified in 4.4.3.

3.8 Accuracy. The average mean radius of the five targets fired shall not exceed 20.0 inches, when tested as specified in 4.4.4.

3.9 Cartridge function. The cartridge shall function when fired in the M39 and M61 guns, when tested as specified in 4.4.6.

3.10 Action time. The action time of the cartridge shall not exceed 4.0 milliseconds in either the ambient phase or the cold phase when tested as specified in 4.4.5.

3.11 Projectile function. The projectile shall detonate high order on impact with the target plate when tested as specified in 4.4.7.

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3.12 Casualties. There shall be no metal parts separation or premature functioning of any projectile nor shall any projectile remain in the weapon bore in any firing test of 4.4. There shall be no misfires in any firing test of 4.4 associated with insufficient, partially burned or contaminated primer composition or propellant.

3.13 Workmanship. The finished cartridge and its components shall be free from cracks, splits, perforations, buckles, folds, laminations or other injurious defects. All parts shall be clean and free of foreign matter. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used. There shall be no contamination in the propellant.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept compete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 The provisions of Specification MIL–G–2550 apply to this specification. Reference will be made to MIL–STD–109 for definition of inspection terms used herein.

4.3 Sampling.

4.3.1 *Lot formation.* A lot shall consist of cartridges loaded by one manufacturer in one unchanged process in accordance with the same drawing and drawing revision, same specification and specification revision, with the following limitations:

(a) Components and subassemblies from one manufacturer and one "Basic Lot" number.

(b) Propellant from one lot only.

4.3.2 *Examination.*

4.3.2.1 Each cartridge or linked belt of the lot shall be examined for the critical defects listed in 4.3.4. Any cartridge or linked belt found to be defective shall be removed from the lot.

4.3.2.2 Unless otherwise specified herein, lot sampling and acceptance criteria will be in accordance with MIL–STD–105 or ORD–M608–11, as appropriate. Judgment for lot acceptance will be based on either the individual defect characteristic or on classes of defect characteristics with individual defect characteristic limitations, as approved by the procuring activity prior to the initiation of production, with the following AQL's assigned (see 6.3):

4.3.2.2.1 Individual defect characteristic basis:

Major Defects	0.25%
Minor Defects	0.40%

4.3.2.2.2 Defect characteristic class basis:

Major Defects	1.50%
Minor Defects	2.50%

Individual defect characteristic AQL's associated with the class basis, will be—

Major Defects	0.40%
Minor Defects	0.65%

4.3.2.3 *Components and subassembies.* Should components and subassemblies be furnished by the Government the contractor shall ascertain by reference to data

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cards, description sheets or other inspection documents, prior to assembly of the cartridges, that such components and subassemblies have been inspected, tested, and accepted in accordance with their respective drawings and specifications. In instances where the contractor also manufactures components or subassemblies for the cartridge, acceptance inspections and tests, including government inspection and tests, will be performed as specified in the pertinent detail specifications for the components or subassemblies.

4.3.3 *Test inspection.* The tests indicated below will be performed in accordance with the methods specified in 4.4. The sample cartridges selected for these tests will be examined for the characteristics listed in 4.3.4.1 and all defective cartridges shall be removed. Test samples will be as shown in Table I. Acceptance criteria are as specified below.

TABLE I

Test	Number of samples
Projectile extraction	10
Velocity and pressure	20
Waterproofness	20
Accuracy	50
Action time (ambient phase)	25
Action time (—65°F. phase)	25
Projectile function	50 (see 6.5)
Function and casualty	400

4.3.3.1 *Projectile extraction.*

4.3.3.1.1 *Acceptance criteria.* For acceptance the sample must meet each of the following criteria:

(a) The sample average range ($\overline{R}$) (see 6.4) may not exceed 813 pounds;

(b) The sample average ($\overline{X}$) plus .85$\overline{R}$ may not exceed the upper specification limit, i.e., $\overline{X} + .85\overline{R} \leqq 2600$ pounds; and

(c) The sample average minus .85$\overline{R}$ should equal or exceed the lower specification limit, i.e., $\overline{X} - 85\overline{R} \geqq 1100$ pounds.

4.3.3.1.2 *Retest.* If the sample fails to meet one or more of the above criteria, a retest of 20 additional cartridges may be conducted using the following criteria:

(a) The sample average range may not exceed 744 pounds;

(b) The sample average plus .89$\overline{R}$ may not exceed the upper specification limit, i.e., $\overline{X} + .89\overline{R} \leqq 2600$ pounds; and

(c) The sample average minus .89$\overline{R}$ should equal or exceed the lower specification limit, i.e., $\overline{X} - .89\overline{R} \geqq 1100$ pounds.

4.3.3.1.3 *Rejection.* If the retest sample fails to meet one or more of the acceptance criteria, the lot will be finally rejected.

4.3.3.1.4 The propellant in the cartridges tested for projectile extraction will be examined for contamination. Presence of any foreign matter in the propellant in any of the cartridges will cause rejection of the lot. (See 6.2).

4.3.3.2 *Firing tests.*

4.3.3.2.1 The lot will be rejected not subject to retest if in any firing test there occurs any of the following defects or malfunctions:

(a) Misfire (associated with (d), (e), (f), 4.3.3.2.9.3).

(b) Projectile remaining in bore.

(c) Metal parts separation (except rotating band separation. See 4.3.3.2.8.1.4).

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(d) Premature functioning of projectile in weapon or in flight before impact.

4.3.3.2.2 *Velocity.* The average velocity may not differ by more than 50 f.p.s. from 3380 f.p.s. nor may the velocity standard deviation of the test sample be more than 40 f.p.s. If the sample fails to meet either or both these criteria, a second sample of 40 cartridges may be tested for the failing parameter(s) (average, standard deviation). Acceptance criteria for retest are the same as those for initial test. The lot will be finally rejected on retest failure.

4.3.3.2.3 *Pressure.*

4.3.3.2.3.1 *Acceptance criterion.* For acceptance, the sample average ($\overline{X}$) plus .84 times the average range ($\overline{R}$) must not exceed 60,500 pounds per square inch (psi), i.e., $\overline{X} + .84\overline{R} \leqq 60{,}500$ psi.

4.3.3.2.3.2 *Retest.* If the sample fails to meet the acceptance criterion a retest may be conducted using an additional sample of 40 cartridges. For acceptance on retest the sample average ($\overline{X}$) plus .91 times the average range ($\overline{R}$) must not exceed 60,500 psi, i.e., $\overline{X} + .91\overline{R} \leqq 60{,}500$ psi.

4.3.3.2.3.3 *Rejection.* The lot will be finally rejected if on retest the sample fails to meet the acceptance criterion.

4.3.3.2.4 *Waterproofness.* The lot will be accepted if the average velocity of the water conditioned sample does not differ by more than 100 f.p.s. from the average velocity of the "dry" sample (4.3.3.2.2). If the average velocity of the water conditioned sample differs by more than 100 f.p.s. from the average velocity of the "dry" sample a retest sample of 40 cartridges will be water conditioned and fired for comparison with an additional sample of 40 "dry" cartridges. The lot will be finally rejected if the average velocity of the water conditioned retest sample differs by more than 100 f.p.s. from the average velocity of the corresponding additional "dry" sample.

4.3.3.2.5 *Accuracy.* If the average mean radius of the 5 targets does not exceed 20 inches the lot will be accepted. If the average mean radius of the 5 targets is greater than 20 inches a retest sample of 100 cartridges (10 targets) will be fired. The lot will be finally rejected if the average mean radius of the retest sample is greater than 20 inches.

4.3.3.2.6 *Action time.* The lot will be rejected, not subject to retest, if any cartridge exhibits an action time greater than 4 milliseconds (.004 seconds) or if the sample average plus 4 standard deviations is greater than 4 milliseconds ($X + 4s > .004$ seconds) in either the ambient or cold temperature phases. The statistics for each phase will be computed individually for the test phase being considered.

4.3.3.2.7 *Projectile function.* The lot will be rejected if five or more projectiles fail to detonate high order on impact with the target. If more than two but less than five projectiles fail to detonate high order, a retest may be fired using 100 cartridges. The lot will be finally rejected if in the combined first test the retest total of five or more projectiles fail to detonate high order.

4.3.3.2.8 *Function and casualty.* The lot will be rejected subject to retest if the number of malfunctions or casualties of any of the types listed below exceeds the "Acc" number but is less than the "Rej" number. Sample size for retest shall be 800 cartridges. The lot will be finally rejected if one or more "Rej" numbers is equaled or exceeded in the first test or in the combined first test and retest.

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TABLE II.

Malfunction	Acc	Rej
Misfire (see 4.3.3.2.9)	—	—
Primer casualties:		
Missing primer or button missing from primer	0	2
Escape of gas around cup or button (see 4.3.3.2.8.1.1)	15	40
Case casualties (see 4.3.3.2.8.1.2):		
Longitudinal split:		
I or S	10	31
J	4	11
K	0	2
L or M	0	1
Circumferential rupture (partial):		
S, J, or K	4	11
L	0	1
Circumferential rupture (complete)	0	1
Detached metal (see 4.3.3.2.8.1.3)	0	1
Failure to extract, including sheared rim	0	1
Failure to chamber	0	1
Rotating band separation (see 4.3.3.2.8.1.4)	0	3

4.3.3.2.8.1 *Firing defect standards.*

4.3.3.2.8.1.1 Gas escape around the periphery of the primer cup or button or through cracked edges of the cup. Suspected leakers as evidenced by smudging of the visible portion of the insulator will be disassembled and any erosion of the insulator will be construed as evidence of a leaker.

4.3.3.2.8.1.2 See Dwg. C7258801 for location of splits and ruptures. If a split or rupture extends into more than one defined position the most severe criterion of Table II will be applied.

4.3.3.2.8.1.3 There must be evidence of detached metal from fired case exterior. The lot will not be penalized for the detachment of metal shavings from the crimped area of the case interior.

4.3.3.2.8.1.4 There must be evidence, through band recovery or hole(s) in the screen(s), of the separation of an entire band or a part of the band having band seat serration imprint. The lot will not be penalized for band slivering or the separation of band fragments showing no evidence of imprint of band seat serrations and having length of 0.5 inches or less. The "Acc" and "Rej" numbers for separated portions of bands showing no evidence of band seat serration imprint and having length greater than 0.5 inches will be 1 and 6, respectively.

4.3.3.2.9 *Misfires.*

4.3.3.2.9.1 All cartridges which misfire will be inspected for:

(a) Primer seating depth improper precluding closing of weapon circuit. This defect wil not be charged against the ammunition lot but will be reported for information.

(b) Metal slivers on head face across insulator.

4.3.3.2.9.2 Following the inspection of 4.3.3.2.9.1 all cartridges which misfire in the M39 or M61 guns will be tested in Rifle, Accuracy and Velocity, 20mm, C7258355 (referenced in Section 5, ORD–M608–14 — Ammunition Ballistic Acceptance Test Methods, Part VIII).

4.3.3.2.9.3 An examination then will be made of each misfiring cartridge, including those from the testing of 4.3.3.2.9.2, covering the following:

(a) Electrical continuity, primed case.

(b) Case vent missing or obstructed.

(c) Absence of primer composition or propellant.

(d) Insufficient primer composition or propellant.

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(e) Partially burned or contaminated primer composition.

(f) Partially burned or contaminated propellant.

4.3.3.2.9.4 The ammunition lot will not be penalized if the misfiring cartridge fires in the test of 4.3.3.2.9.2 or if none of the causes of 4.3.3.2.9.3 can be assigned.

4.3.3.2.9.5 The lot will be rejected not subject to retest if cause 4.3.3.2.9.3 (d), (e), or (f) is found. The "Acc" number shall be zero and the "Rej" number 3 for those misfires associated with causes 4.3.3.2.9.1 (b) or 4.3.3.2.9.3 (a), (b), or (c).

4.3.3.2.10 *Defective weapons.*

4.3.3.2.10.1 If any firing defect is found to have been caused by the weapon used, such firing defect will not be counted against the ammunition, but the weapon will be corrected or replaced and the test repeated. This applies also to any retest of ammunition.

4.3.3.2.10.2 If the cyclic rate of fire in any one burst fired on either the first test or retest in the M39 Gun falls below 1400 rounds per minute, or if the rate of fire falls below 5500 rounds per minute in the M61 Gun, the following action will be taken:

4.3.3.2.10.3 Failure to meet the minimum cyclic rate in the M61 Gun will cause suspension of the test and nullification of the test results in the failing burst except for the defects listed in 4.3.3.2.8 having an acceptance number of zero (0). The test will be continued after repair or replacement of the gun.

4.3.3.2.10.4 On failure of a burst to meet the minimum cyclic rate in the M39 Gun, the following burst will be fired in a second gun. If any burst fails to meet the required cyclic rate in the second gun, that gun will be fire-checked with a 50 round burst of ammunition from a previously accepted lot. Failure to meet the cyclic rate with the accepted ammunition, indicative of gun malfunction, will nullify the results of the test ammunition bursts, except for the defects listed in 4.3.3.2.8, and the malfunctioning weapon(s) will be repaired or replaced. The test then will be continued from the point at which the failure occurred. All test results to that point will be considered valid and will be considered along with the results of subsequent firing necessary to attain a sample of 200 test cartridges. If the burst following the failing burst, or the burst of cartridges from a previously accepted lot, functions the second gun at the required minimum cyclic rate, indicative of ammunition failure in the failing burst, the ammunition lot will be finally rejected and referred to the responsible Procuring activity for disposition.

4.3.3.2.11 *Unusual firing defects.* Any defects not covered by this specification that are found by the Government inspector during the acceptance tests will be recorded on the acceptance report and brought to the attention of the contracting officer. The contracting officer will report the defects to the responsible procuring activity for action along with documents of the manufacturer when, in the opinion of the contracting officer, the defects indicate that the ammunition is unsuited for the purpose intended. Final acceptance of the lot will be held in abeyance pending decision by the responsible procuring activity.

4.3.4 *Classification of defects.*

4.3.4.1 *Cartridge assembly.* (See Drawing D7553814.)

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Category and defects	Method of inspection
Crtical	
1. Crack or split in shell body	Visual
2. Split or perforated case	Visual
Major	
101. Loose primer	Manual (see 4.3.5.1)
102. Depth of primer seating	Gage
103. Overall length	Gage
104. Improper or missing crimp (case-projectile)	Visual (see 4.3.5.2)
105. Profile and alignment, max.	Gage
106. Corrosion	Visual
107. Head configuration	Visual (see 4.3.5.3)
108. Metal slivers on head face	Visual
109. Fuze or fuze cover loose	Test (see 4.3.5.4)
110. Crack, split or dent in fuze cover	Visual
Minor	
201. Foreign matter	Visual
202. Gap between case and rotating band	Visual-Gage (see 4.3.5.5)
203. Marking incorrect or unidentifiable	Visual
204. Gap between fuze and shell	Gage (see 4.3.5.6)
205. Protective coating damaged or incomplete	Visual (see 4.3.5.7)
206. Waterproofing missing around primer	Visual
207. Projectile fails to withstand torque test	Testing device
208. Scratch, dent, bulge, wrinkle or fold in case	Visual
209. Lacquer on primer button	Visual

4.3.4.2 Metallic link belt taken from M2A1 or T46 box (See Drawing FD 20356, FD 20357, FD 20380 or FD 20381 as applicable, covering a detail of FD 20354, or Drawing FD 8650, FD 8651 or FD 8652 as applicable, covering a detail of FD 20421).

Category and defects	Method of inspection
Critical	
1. Belt flexibility (M17 linked cartridges)	Manual (see 4.3.5.9)
Major	
101. Stretched, frozen or broken belt	Visual (see 4.3.5.10)
102. Distorted link	Visual
103. Misalignment of cartridge in linked belt	Visual
Minor	
201. Improper packing of belt in container	Visual
202. Incorrect type of ammunition	Visual
203. Foreign matter, oil or grease, other than normally applied	Visual
204. Defective protective finish or rust on link	Visual
205. Improper number of cartridges in belt	Visual
206. Missing or improper fillers or separators	Visual

4.3.4.3 Bulk pack taken from M2A1 box. (See Dwg. FD 20355 covering a detail of FD 20353.)

Category and defects	Method of inspection
Critical	
None defined.	
Major	
None defined.	
Minor	
201. Incorrect number of cartridges in container	Visual
202. Missing or improper fillers, or tubes	Visual
203. Improper packing of cartridges in container	Visual
204. Incorrect type of ammunition	Visual

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4.3.4.4 *Closed container.* (See Dwg. FC 8524 or FC 8523, covering details of FD 20354 and FD 20353, respectively, or Dwg. FD 20421, as applicable.)

Category and defects	*Method of inspection*
Critical	
None defined.	
Major	
101. Incorrectly closed and secured box	Visual
102. Missing or unserviceable end handle or carrying handle	Visual
103. Missing, incorrect or unidentifiable lot number (or FSN on T46 box)	Visual
Minor	
201. Missing or defective component other than end handle or carrying handle	Visual
202. Burrs or sharp edges	Visual
203. Severe dents, buckles or other damage	Visual
204. Scratches or abrasions exposing bare metal	Visual
205. Marking other than ammunition lot number (or FSN on T46 box) missing, incomplete or unidentifiable	Visual
206. Incomplete paint coverage	Visual
207. Marking touched up or repaired with nonpermissible material (crayon, chalk, etc.)	Visual

4.3.4.5 Packed wirebound box (see Dwg. FD 20353 or FD 20354, as applicable).

Category and defects	*Method of inspection*
Critical	
Non defined.	
Major	
101. FSN or lot number missing, incomplete, unidentifiable or incorrect at any location on box	Visual (see 4.3.5.8)
102. Missing batten or cleat	Visual
103. Filler or separator missing	Visual
104. Packed ammunition not in agreement with FSN	Visual
Minor	
201. Marking other than FSN or lot number missing, incomplete, unidentifiable or incorrect	Visual
202. Marking touched up or repaired with nonpermissible material (crayon, chalk, etc.)	Visual

4.3.5 *Defect standards.*

4.3.5.1 A primer which moves with respect to the case, under ordinary finger pressure, will be classed a defect.

4.3.5.2 Any cartridge having other than a smooth continuous crimp for 360° on the case periphery or having misplaced or improperly located crimp not filling the projectile crimp groove will be classed defective.

4.3.5.3 The cartridge will be classed defective if there are any head defects such as bent head, head out of round, "round head" or "beveled underside of head."

4.3.5.4 A cartridge will be classed defective when the fuze turns, with respect to the shell, upon application of 10 inch-pounds torque applied in a counterclockwise direction.

4.3.5.5 If a gap of $\frac{1}{32}$ inch or more exists between the cartridge case mouth and the rear edge of the rotating band the cartridge will be classed defective.

4.3.5.6 Any gap, in excess of .005 inch. at any point of the circumference will be classed a defect.

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4.3.5.7 The cartridge will be classed defective if more than 1/32 square inch of the projectile surface is unpainted or if there is evidence of the action of a corrosive agent which has lessened the effectiveness of the protective coating.

4.3.5.8 The FSN will include the DOD code.

4.3.5.9 The belt shall hinge freely and fold over smoothly without kinking when the belt is pulled over itself until belt is completely reversed. The procedure will then be repeated after the belt has been reversed to assure full motion of the belt when flexed from either side, in either direction. A minimum of 25 cartridges per belt will be used in this test. If packing instructions require belts of length greater than 25 cartridges, the connecting links used to lengthen belts will be flexed after assembly in both directions to assure free hinging.

4.3.5.10 This quality characteristic applies to M12, M14 and M17 linked belts for stretched or broken belts and to M12 and M14 for frozen belts.

4.3.6 *Examination equipment.*

4.3.6.1 *Government verification inspection.* Inspection willl be accomplished with the use of gages and equipment specified in the "Index to Lists of Inspection Gages and Parts Gaged On: Cartridge, HE1, 20MM, M56A1," GL 7555938.

4.4 Test methods.

4.4.1 *Electrical resistance, primed cases.* The primed cases will be tested for electrical resistance subsequent to crimping the primers into the cases and application of primer waterproofing, and prior to assembly of the primed cases into cartridges. The testing will be accomplished with circuits specified on Dwg. D7553513.

4.4.2 *Projectile extraction.* The sample cartridges will be tested as specified in Section II of ORD–M608–14 — Ammunition Ballistic Acceptance Test Methods, Part VIII. The statistics necessary for testing of results against acceptance criteria will be computed.

4.4.3 *Velocity, pressure and waterproofness.* The tests will be conducted as prescribed in Section 3 of ORD–M608–14 — Ammunition Ballistic Acceptance Test Methods, Part VIII. Statistics necessary for testing of results against acceptance criteria will be computed.

4.4.4 *Accuracy.* The method of test and measurement of targets will be as prescribed in Section 5 of ORD–M608–14 — Ammunition Ballistic Acceptance Test Methods, Part VIII. The range may be from 100 to 600 yards in length. Ten shot targets will be used.

4.4.5 *Action time.* The method of test will be as prescribed in Section 6 of ORD–M608–14 — Ammunition Ballistic Acceptance Test Methods, Part VIII. Statistics necessary for testing of results against the acceptance criterion will be computed. Of the test samples, 25 cartridges will be fired at ambient temperature and 25 cartridges will be conditioned and fired at —65°F.

4.4.6 *Function and casualty.* Two hundred (200) cartridges will be fired fully automatic in bursts of 50 rounds in an M39 gun. Two hundred (200) cartridges will be fired fully automatic in bursts of 50 rounds at a cyclic rate of not less than 5500 rounds per minute in an M61 gun. The method of test will be as specified in Section 4 of ORD–M608–14 Ammunition Ballistic Acceptance Test Methods, Part VIII. The projectiles fired in this test will be observed for a distance of 600 feet for premature bursts. The gun barrel(s) will be at ambient temperature at the beginning of the test and will be cooled to ambient temperature between bursts.

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4.4.7 *Projectile function.* The test procedure will be as prescribed in Section 7 of ORD–M608–14 — Ammunition Ballistic Acceptance Test Methods, Part VIII. Impact of the projectile will be against an aluminum target (2024T4, T3 or T36) of .040″ nominal thickness placed 200 yards, ± 25 yards, from the gun muzzle normal to the line of fire. Observation will be made from a position to the side of and slightly down range from the target to assure an accurate determination of the point of functioning and character of detonation.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A and B.* Ammunition shall be packaged as specified on Dwgs. FD 20353, FD 20354, or FD 20421, as applicable, or as specified by the procuring agency or the contracting officer.

5.1.2 *Level C.* Ammunition shall be bulk packed as specified by the procuring agency or the contracting officer.

5.2 Packing.

5.2.1 *Level A and B.* Ammunition preserved and packaged in Box, Ammunition, Caliber .50 M2A1 shall be overpacked in accordance with Dwg. FD 20353 or FD 20354, as applicable, or as specified by the procuring agency or the contracting officer.

5.2.2 *Level C.* As specified by the procuring agency or the contracting olcer.

5.3 Marking.

5.3.1 *Levels A, B or C.* Packaged and packed ammunition shall be marked in accordance with Dwgs. FD 20421, FD 20353 or FD 20354, as applicable, or as directed by the procuring agency or contracting officer. All marking shall be in accordance with ICC Regulation 49 CFR 71–78, and with MIL–STD–129, to the extent applicable.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Place of inspection, if not at place of manufacture.

(c) Detailed packaging and packing required.

(d) Marking required on shipping containers.

6.2 Particular care should be exercised to avoid contamination of primers and propellant powder by oil, grease, and other foreign material, since such contamination may increase action time and make the ammunition unsuitable for aircraft guns.

6.3 AQL's. The optional use of AQL values for either individual defects or classes of defects, as approved by the procuring activity prior to the initiation of production, is authorized to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the process.

6.4 Where computation of a sample standard deviation is prescribed for determination of lot acceptance, any accepted method of computation may be used and the data may be handled without regard to order. However, where a range plan is prescribed, the individual values will be recorded in subgroups of 5 in the order in which they are observed. The ranges for the sub-groups will be added and the sum of the ranges will be divided by the number of sub-groups to determine the average range ($\bar{R}$).

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6.5 After all tests have been completed and final acceptance of the cartridge lot is accomplished, a random sample of 400 cartridges from the lot will be forwarded to a test facility to be designated by the procuring activity. These cartridges will be fired to determine if spontaneous projectile functions occur at an altitude of 40,000 feet plus or minus 2,000 feet, with an aircraft speed of MACH 1 plus or minus MACH .05. Test results will be furnished to the procuring activity.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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